

Notes on lecture 12

Jiten Chandra Kalita and Shyamashree Upadhyay

1 Properties of double integrals

Listed below (Properties 6,7, and 8) are some properties of double integrals. Earlier, we had seen these properties for a rectangular region R . Here they are listed for a general region D .

$$\boxed{6} \quad \iint_D [f(x, y) + g(x, y)] dA = \iint_D f(x, y) dA + \iint_D g(x, y) dA$$

$$\boxed{7} \quad \iint_D c f(x, y) dA = c \iint_D f(x, y) dA$$

If $f(x, y) \geq g(x, y)$ for all (x, y) in D , then

$$\boxed{8} \quad \iint_D f(x, y) dA \geq \iint_D g(x, y) dA$$

Remark 1.0.1. The given region D may not be beautifully shaped so that one can easily convert it into a type I or a type II region. There are regions which can't be expressed directly as type I or type II regions. But if we examine the situation closely, we will see that any region D can be expressed as a union of type I or type II regions.

The next property of double integrals is similar to the property of single integrals given by the equation $\int_a^b f(x) dx = \int_a^c f(x) dx + \int_c^b f(x) dx$.

If $D = D_1 \cup D_2$, where D_1 and D_2 don't overlap except perhaps on their boundaries (see Figure 17), then

9

$$\iint_D f(x, y) dA = \iint_{D_1} f(x, y) dA + \iint_{D_2} f(x, y) dA$$

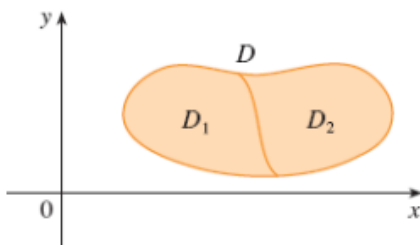


Figure 1.0.1: Figure 17

This formula (Formula 9) is going to help us in decomposing a domain D (which is not explicitly a type I or type II region) as a union of type I or type II regions. Please see Figure 1.0.2 for an illustration.

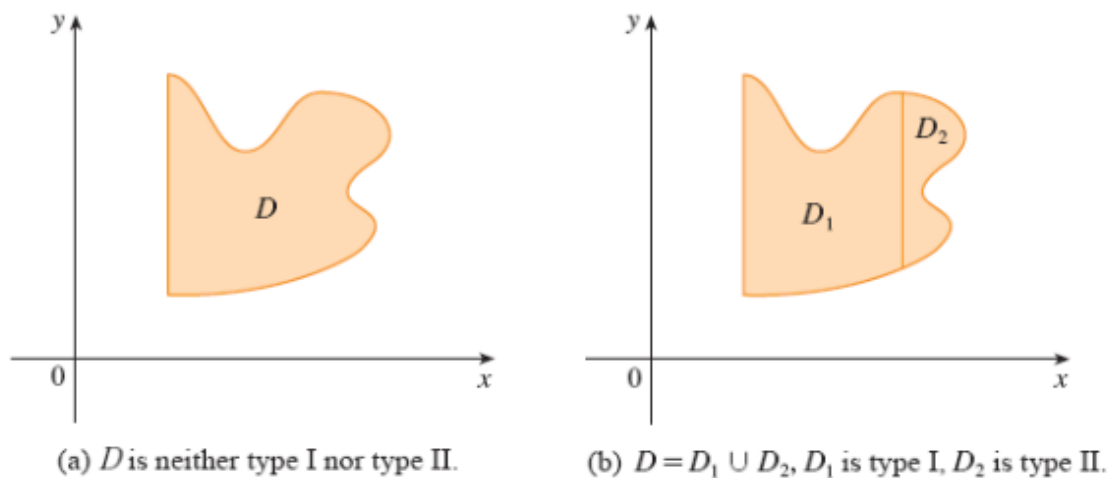


Figure 1.0.2: D as a union of type I and type II regions

Example 1.0.2. Express D as a union of regions of type I or type II and evaluate the integral $\int_D \int y dA$, where D is the shaded portion as given in Figure 1.0.3. Obviously, D is neither of type I nor of type II. Let us divide

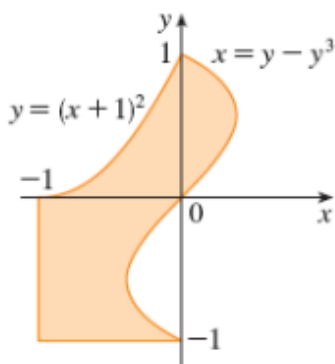


Figure 1.0.3: Figure for Example 1.0.2

the region D into 3 parts D_1, D_2, D_3 , where

$$D_1 = \{(x, y) | -1 \leq x \leq 0, 0 \leq y \leq (x+1)^2\}$$

$$D_2 = \{(x, y) | 0 \leq y \leq 1, 0 \leq x \leq y - y^3\}$$

and

$$D_3 = \{(x, y) | -1 \leq y \leq 0, -1 \leq x \leq y - y^3\}.$$

Clearly, D_1 is a type I region, and D_2, D_3 are both of type II. Then we have

$$\int_D \int y dA = \int_{D_1} \int y dA + \int_{D_2} \int y dA + \int_{D_3} \int y dA.$$

The rest of the calculation is left to the reader as an exercise. $\square$

The next property of integrals says that if we integrate the constant function $f(x, y) = 1$ over a region D , we get the area of D :

(10)

$$\iint_D 1 dA = A(D)$$

(11) If $m \leq f(x, y) \leq M$ for all (x, y) in D , then

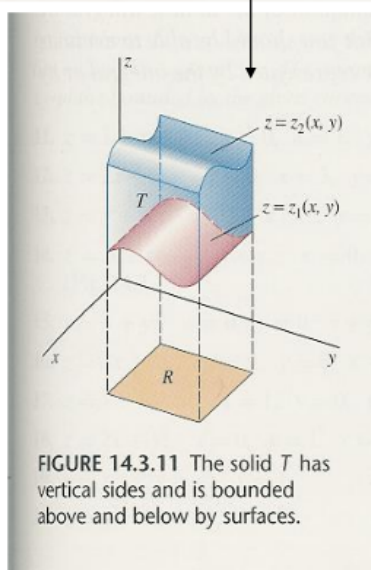
$$mA(D) \leq \iint_D f(x, y) dA \leq MA(D)$$

1.1 Volume between two surfaces

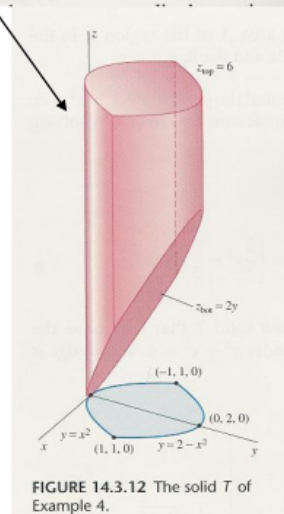
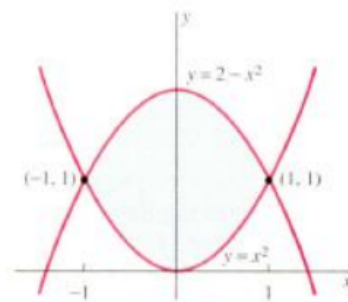
Volume Between Two Surfaces

Suppose now that the solid region T lies above the plane region R , as before, but *between* the surfaces $z = z_1(x, y)$ and $z = z_2(x, y)$, where $z_1(x, y) \leq z_2(x, y)$ for all (x, y) in R (Fig. 14.3.11). Then we get the volume V of T by subtracting the volume below $z = z_1(x, y)$ from the volume below $z = z_2(x, y)$, so

$$V = \iint_R [z_2(x, y) - z_1(x, y)] dA. \quad (5)$$



EXAMPLE 4 Find the volume V of the solid T bounded by the planes $z = 6$ and $z = 2y$ and by the parabolic cylinders $y = x^2$ and $y = 2 - x^2$. This solid is sketched in Fig. 14.3.12.



Remark 1.1.1. Formula 5 above is given for a rectangular region R . But it can be easily extended to any general region.

2 Relation between cartesian and polar coordinates

Till now, we have studied double integrals. We talked about evaluating the integral $\int_D \int f(x, y) dA$, where D was a general region. So far, in whatever

problems we have considered, we have defined the region D in terms of cartesian coordinate system. But in many problems, the region D may not be easily recognizable in terms of the cartesian coordinate system. For example, suppose we have a domain R of the shape of a circular disk (as given in Figure 2.0.1) In terms of cartesian coordinates,

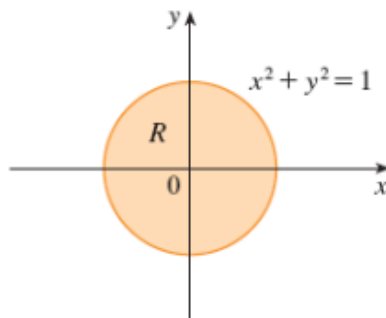


Figure 2.0.1: A circular disk

$$R = \{(x, y) | -1 \leq x \leq 1, -\sqrt{1-x^2} \leq y \leq \sqrt{1-x^2}\}.$$

The above expression for R (in terms of cartesian coordinates) is complicated enough. We will have to work with some upper and lower limits of integration like $\pm\sqrt{1-x^2}$, which may be cumbersome. Also, depending upon the integrand $f(x, y)$, the computation is going to be simpler or more complicated.

We may also have something like the shaded portion R in Figure 2.0.2 as our domain.

In such cases, it is easier to recognize the region in terms of polar coordinates. We are now going to find out double integrals $\int_R \int f(x, y) dA$ over a domain R , which is expressible in terms of polar coordinates. We also know that if we have a situation as in Figure 2.0.3, then the relation between the polar coordinates (r, θ) and the cartesian coordinates (x, y) is given by:

$$x = r \cos \theta, \quad y = r \sin \theta, \quad r^2 = x^2 + y^2.$$

Now we will try to find out double integrals over regions which are expressible in terms of polar coordinates.

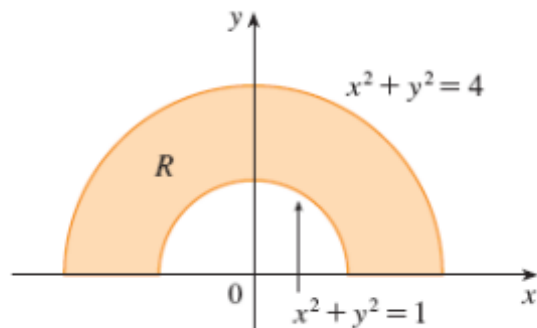


Figure 2.0.2: An annulus

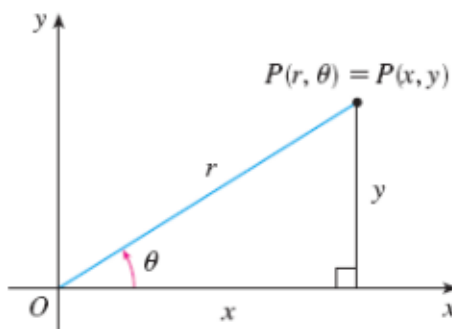


Figure 2.0.3: Relation between cartesian and polar coordinates

3 Double integrals in polar coordinates

In cartesian coordinate system, we started with a rectangular region. Likewise, here we are going to start with a “polar rectangular region”. A **polar rectangular region** is a region R of the form

$$R = \{(r, \theta) | a \leq r \leq b, \alpha \leq \theta \leq \beta\}.$$

Please see Figure 3.0.1 for an illustration. We are going to divide this polar rectangle into some sub-polar rectangles. For this, we divide the interval $[a, b]$ as

$$r_0 = a, r_1, \dots, r_n = b.$$

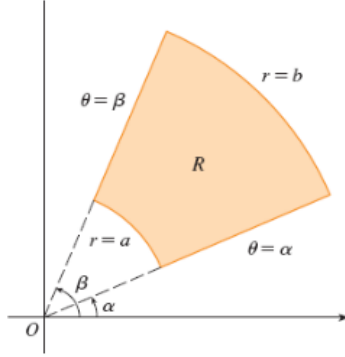


Figure 3.0.1: A polar rectangular region

And we divide the interval $[\alpha, \beta]$ as

$$\theta_0 = \alpha, \theta_1, \dots, \theta_m = \beta.$$

These intervals need not be equally spaced. Suppose we now consider a sub-polar rectangle

$$R_{ij} = \{(r, \theta) | r_{i-1} \leq r \leq r_j, \theta_{j-1} \leq \theta \leq \theta_j\}.$$

Please see Figure 3.0.2 for an illustration. Then we consider a point (r_i^*, θ_j^*) in R_{ij} . We may take

$$r_i^* = \frac{1}{2}(r_{i-1} + r_i)$$

and

$$\theta_j^* = \frac{1}{2}(\theta_{j-1} + \theta_j).$$

If ΔA_{ij} is the area of R_{ij} , then

ΔA_{ij} = the area of the bigger polar triangle—the area of the smaller polar triangle.

The yellow colored shaded portion in Figure 3.0.3 is the area ΔA_{ij} . Since the area of a polar triangle with radius r and angle θ is given by $\frac{1}{2}r^2\theta$, it follows that

$$\Delta A_{ij} = \frac{1}{2} (r_i^2 - r_{i-1}^2) \Delta \theta_j,$$

where $\Delta \theta_j = \theta_j - \theta_{j-1}$. In the pictures and in Stewart's book, sometimes the notation $\Delta \theta$ is used for $\Delta \theta_j$ and ΔA_i for ΔA_{ij} . Then we have

$$\Delta A_{ij} = \frac{1}{2}(r_i + r_{i-1})(r_i - r_{i-1})\Delta \theta_j = r_i^* \Delta r_i \Delta \theta_j,$$

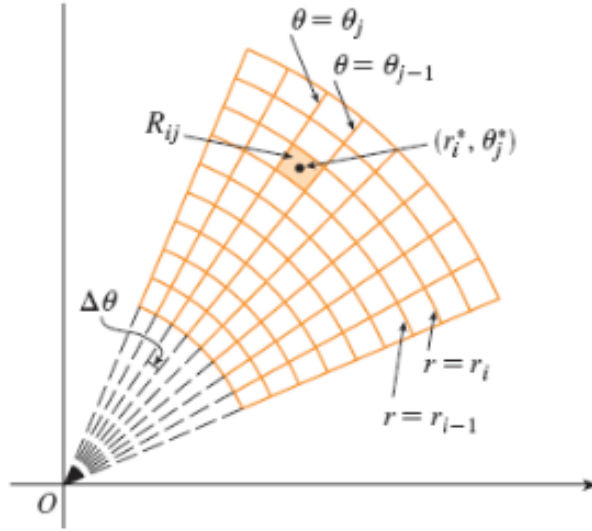


Figure 3.0.2: Dividing into sub-polar rectangles

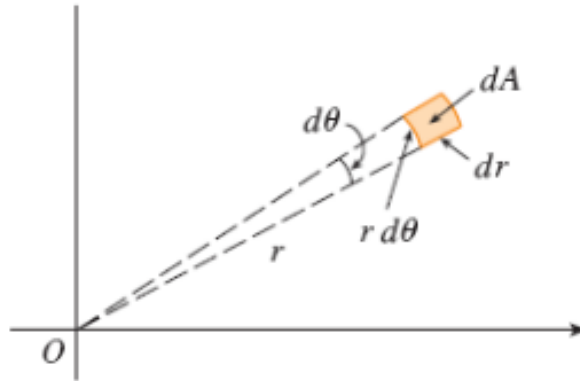


Figure 3.0.3: The shaded portion is the area ΔA_{ij}

where $\Delta r_i = r_i - r_{i-1}$.

Now we will consider $\int_R \int f(x, y) dA$, where R was given by

$$R = \{(r, \theta) | a \leq r \leq b, \alpha \leq \theta \leq \beta\}.$$

Then we will be firstly intersted in the product

$$f(r_i^* \cos \theta_j^*, r_i^* \sin \theta_j^*) \Delta A_{ij}.$$

This expression is going to represent the volume of the column that lies above the sub-polar rectangle R_{ij} and below the portion of the surface given by the function $f(x, y)$ (provided we assume that $f(x, y) \geq 0$ on R).

Definition 3.0.1. The integral $\int_R \int f(x, y) dA$ is defined as

$$\int_R \int f(x, y) dA = \lim_{m, n \rightarrow \infty} \sum_{i=1}^n \sum_{j=1}^m f(r_i^* \cos \theta_j^*, r_i^* \sin \theta_j^*) \Delta A_{ij}.$$

We know that $\Delta A_{ij} = r_i^* \Delta r_i \Delta \theta_j$. That's why $\int_R \int f(x, y) dA$ is going to become

$$\int_{\alpha}^{\beta} \int_a^b f(r \cos \theta, r \sin \theta) r dr d\theta.$$

□

Remark 3.0.2. In the above discussion, we must have $0 \leq \beta - \alpha \leq 2\pi$.

Example 3.0.3. Find the volume of the solid bounded by the plane $z = 0$ and the paraboloid $z = 1 - x^2 - y^2$. Figure 3.0.4 gives a picture of the solid. This volume is equal to the integral $\int_D \int (1 - x^2 - y^2) dA$, where D is the disk given by

$$D = \{(x, y) | x^2 + y^2 \leq 1\}.$$

In polar coordinates, D is given by

$$D = \{(r, \theta) | 0 \leq r \leq 1, 0 \leq \theta \leq 2\pi\}.$$

Then

$$\int_D \int (1 - x^2 - y^2) dA = \int_{\theta=0}^{2\pi} \int_{r=0}^1 (1 - r^2) r dr d\theta.$$

In the last lecture, we forgot to mention about a theorem, which states that If $f(x, y) = f_1(x)f_2(y)$, then

$$\int_{x=a}^b \int_{y=c}^d f(x, y) dy dx = \int_a^b f_1(x) dx \int_c^d f_2(y) dy.$$

Following this theorem, we get that

$$\int_D \int (1 - x^2 - y^2) dA = \int_0^{2\pi} 1 d\theta \int_0^1 (r - r^3) dr = 2\pi \left(\frac{1}{2} - \frac{1}{4} \right) = \frac{\pi}{2}.$$

□

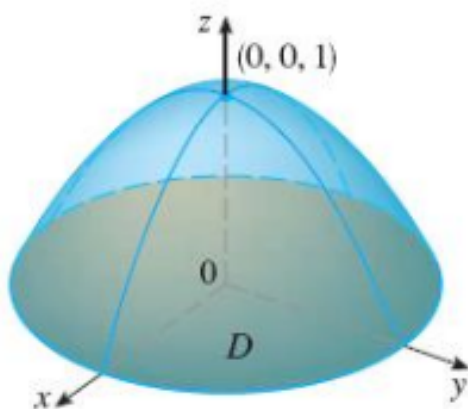


Figure 3.0.4: Figure for Example 3.0.3

3.1 Type I and II regions in polar coordinates

In polar coordinate system also, we can have type I and type II regions. A **type I polar region** is of the form

$$D = \{(r, \theta) | \alpha \leq \theta \leq \beta, h_1(\theta) \leq r \leq h_2(\theta)\}.$$

Figure 3.1.1 depicts a type I polar region.

If f is continuous on a polar region D of the form

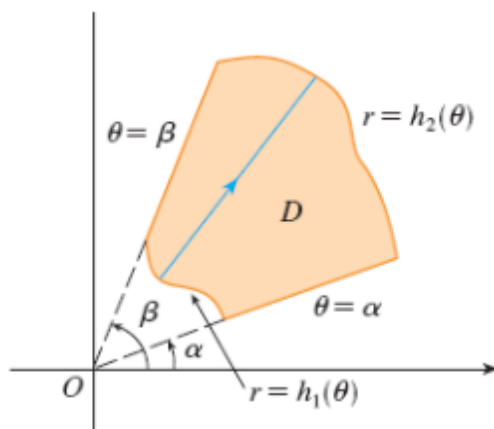


Figure 3.1.1: A type I polar region

$$D = \{(r, \theta) | \alpha \leq \theta \leq \beta, h_1(\theta) \leq r \leq h_2(\theta)\},$$

then

$$\int_D \int f(x, y) dA = \int_{\alpha}^{\beta} \int_{h_1(\theta)}^{h_2(\theta)} f(r \cos \theta, r \sin \theta) r dr d\theta.$$

In this integral on the right hand side, we will have to first integrate with respect to r .

Similarly, a **type II polar region** is a region D of the form

$$D = \{(r, \theta) | a \leq r \leq b, g_1(r) \leq \theta \leq g_2(r)\}.$$

Figure 3.1.2 depicts a type II polar region.

If f is continuous on a polar region D of the form

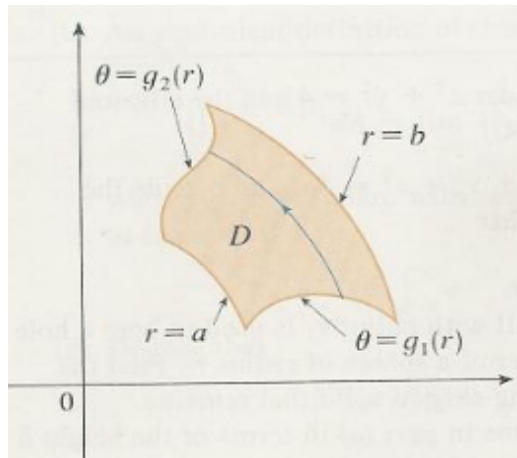


Figure 3.1.2: A type II polar region

$$D = \{(r, \theta) | a \leq r \leq b, g_1(r) \leq \theta \leq g_2(r)\},$$

then

$$\int_D \int f(x, y) dA = \int_a^b \int_{g_1(r)}^{g_2(r)} f(r \cos \theta, r \sin \theta) r d\theta dr.$$

In this integral on the right hand side, we will have to first integrate with respect to θ .

Example 3.1.1. Find the volume of the solid that lies under the paraboloid $z = x^2 + y^2$, above the xy -plane, and inside the cylinder $x^2 + y^2 = 2x$. The right hand side picture in Figure 3.1.3 depicts the solid and the left hand side picture in Figure 3.1.3 depicts the region D on which we have to integrate the integrand.

To compute this volume, we will have to integrate $x^2 + y^2$ over the region

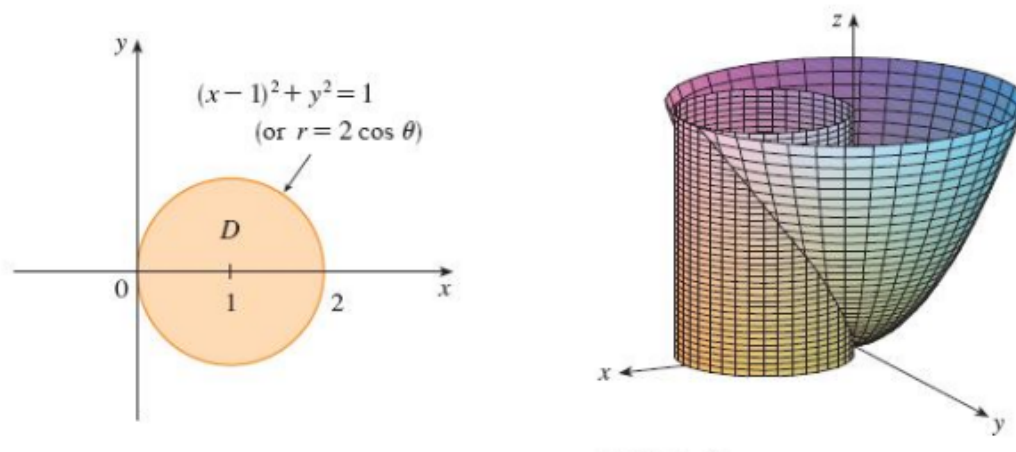


Figure 3.1.3: Figure for Example 3.1.1

$$D = \{(x, y) | (x - 1)^2 + y^2 \leq 1\}.$$

Putting $x = r \cos \theta$ and $y = r \sin \theta$ in the equation $(x - 1)^2 + y^2 = 1$, we get

$$r^2 \cos^2 \theta - 2r \cos \theta + 1 + r^2 \sin^2 \theta = 1.$$

Or in other words,

$$r^2 - 2r \cos \theta = 0 \Rightarrow r = 0, 2 \cos \theta.$$

This implies that r ranges from 0 to $2 \cos \theta$. And clearly, θ ranges from $-\frac{\pi}{2}$ to $\frac{\pi}{2}$. So we have

$$D = \{(r, \theta) | -\frac{\pi}{2} \leq \theta \leq \frac{\pi}{2}, 0 \leq r \leq 2 \cos \theta\}.$$

That is, D is a type I polar region. Then

$$\int_D \int (x^2 + y^2) dA = \int_{\theta=-\frac{\pi}{2}}^{\frac{\pi}{2}} \int_{r=0}^{2 \cos \theta} r^2 r dr d\theta.$$

In this integral, we will have to first integrate with respect to r . The rest of the calculation is left to the reader as an **exercise**. $\square$

3.2 Application of double integrals

The reader is urged to read pages 19-22 of the lecture slide 12 for this.

4 Surface area

In earlier lectures, we already studied surface area for surfaces of revolution about an axis. But now, we will be interested in surface area of the graph of a function of 2 variables, and for simplicity, we are going to assume that $f(x, y) \geq 0$.

Remark 4.0.1. We have assumed for simplicity that $f(x, y) \geq 0$. But this doesn't mean that we can't work with functions $f(x, y)$ which are NOT so. In the general case, we will have to divide the domain of definition of the function f in such a way that either we are in the case $f(x, y) \geq 0$ or in the case $f(x, y) \leq 0$.

Here, we are going to talk about surface area of the graph of a function whose domain of definition is a rectangle, say

$$R = [a, b] \times [c, d].$$

Once we know how to work it out for a rectangular domain, then we already know how to extend that idea to a general domain.

Suppose $f(x, y) \geq 0$ for all $(x, y) \in R$. Let us divide the interval $[a, b]$ into n many subintervals by

$$a = x_0 < x_1 < \cdots < x_n = b.$$

Similarly, let us divide the interval $[c, d]$ into m many subintervals by

$$c = y_0 < y_1 < \cdots < y_m = d.$$

By doing so, we have divided R into mn -many sub-rectangles. Consider the sub-rectangle

$$R_{ij} = [x_{i-1}, x_i] \times [y_{j-1}, y_j].$$

Let us now consider the surface over R_{ij} , that is, the graph of the function for this sub-rectangle. Let us also consider the point (x_{i-1}, y_{j-1}) in R_{ij} . Let ΔS_{ij} denote the surface area of the surface lying above R_{ij} . Consider the tangent plane ΔT_{ij} at the point (x_{i-1}, y_{j-1}) to the surface, that is above the

sub-rectangle. Let the area of the tangent plane be ΔA_{ij} . Let $\Delta x_i = x_i - x_{i-1}$ and $\Delta y_j = y_j - y_{j-1}$. If we are able to find out the areas of all ΔT_{ij} , then sum them up, and then let $m, n \rightarrow \infty$, we are going to get the area of the entire surface. So it all boils down to finding out the area of ΔT_{ij} . The left hand side picture in Figure 4.0.1 explains this phenomenon.

Let us consider the right hand side picture in Figure 4.0.1. Let us consider

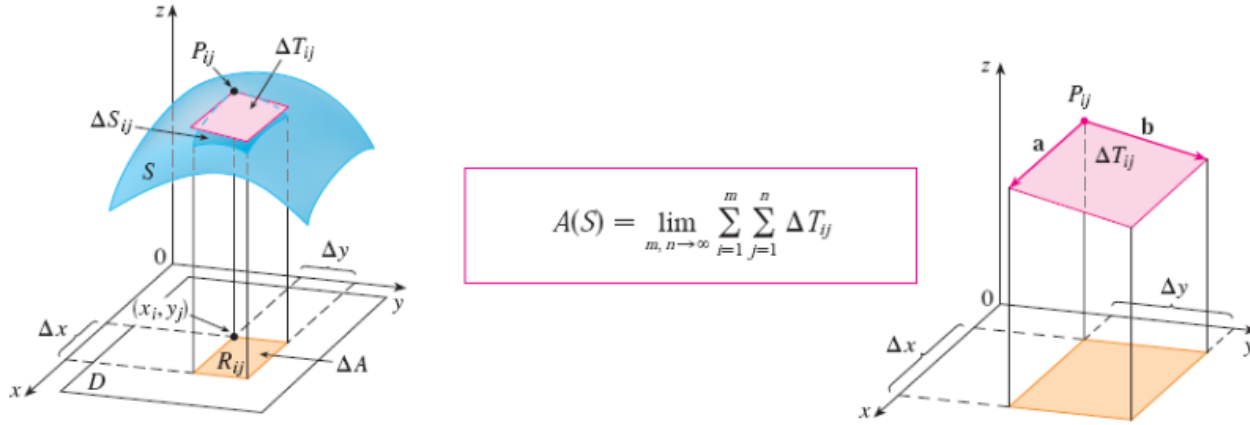


Figure 4.0.1: Surface area

the vectors $\vec{a}$ and $\vec{b}$ (as shown in Figure 4.0.1). Then the area of the plane ΔT_{ij} equals the area of the parallelogram created by the sides $\vec{a}$ and $\vec{b}$, which in turn is equal to $|\vec{a} \times \vec{b}|$.

Now, we can also see that the vector $\vec{a}$ is going to lie in the xz -plane. Similarly, the vector $\vec{b}$ is going to lie in the yz -plane.

The x -component of the vector $\vec{a}$ is Δx ($= \Delta x_i$). Let the z -component of the vector $\vec{a}$ be denoted by Δz . That is,

$$\vec{a} = \Delta x \hat{i} + \Delta z \hat{k}.$$

Let θ be the angle that the vector $\vec{a}$ makes with the positive direction of the x -axis. Then we have

$$\frac{\Delta z}{\Delta x} = \tan \theta.$$

Remember that the vector $\vec{a}$ is nothing but a tangent to the curve which has been obtained by intersecting the surface with the plane $y = y_{j-1}$. Now,

from the geometrical interpretation of partial derivatives, we have:

$$\frac{\Delta z}{\Delta x} = \tan \theta = f_x(x_{i-1}, y_{j-1}).$$

This implies that

$$\Delta z = f_x(x_{i-1}, y_{j-1})\Delta x.$$

Remark 4.0.2. In Stewart's book as well as in Figure 4.0.1, the point (x_i, y_j) is taken instead of (x_{i-1}, y_{j-1}) . But there should not be any problem due to this as $m, n \rightarrow \infty$.

So we now have

$$\vec{a} = \Delta x \hat{i} + f_x(x_{i-1}, y_{j-1})\Delta x \hat{k}.$$

Similarly,

$$\vec{b} = \Delta y \hat{j} + f_y(x_{i-1}, y_{j-1})\Delta y \hat{k},$$

where $\Delta y = \Delta y_j$. Then we have

$$\vec{a} \times \vec{b} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \Delta x & 0 & f_x \Delta x \\ 0 & \Delta y & f_y \Delta y \end{vmatrix},$$

where f_x means $f_x(x_{i-1}, y_{j-1})$ and f_y means $f_y(x_{i-1}, y_{j-1})$.

So we get

$$\vec{a} \times \vec{b} = (-f_x \hat{i} - f_y \hat{j} + \hat{k}) \Delta x \Delta y.$$

This implies that

$$|\vec{a} \times \vec{b}| = \sqrt{1 + f_x^2 + f_y^2} (\Delta x \Delta y).$$

$$\Rightarrow \Delta A_{ij} = \sqrt{1 + (f_x(x_{i-1}, y_{j-1}))^2 + (f_y(x_{i-1}, y_{j-1}))^2} \Delta x \Delta y.$$

Hence

$$\begin{aligned} \lim_{m, n \rightarrow \infty} \sum_{i=1}^n \sum_{j=1}^m \Delta A_{ij} &= \lim_{m, n \rightarrow \infty} \sum_{i=1}^n \sum_{j=1}^m \sqrt{1 + (f_x(x_{i-1}, y_{j-1}))^2 + (f_y(x_{i-1}, y_{j-1}))^2} \Delta x \Delta y. \\ &= \int_R \int \sqrt{1 + f_x^2 + f_y^2} \, dA. \end{aligned}$$

This is the formula for surface area.

Remark 4.0.3. f is such that the first order partial derivatives not only just exist over R , but they are also continuous.

Example 4.0.4. Find the surface area of the part of the surface $z = x^2 + 2y$ that lies above the triangular region T in the xy -plane with vertices $(0, 0)$, $(1, 0)$, $(1, 1)$. The right hand side picture in Figure 4.0.2 shows the surface and the left hand side picture in the same figure shows the region T .

Let $f(x, y) = x^2 + 2y$. Then $f_x = 2x$ and $f_y = 2$. So we have $\sqrt{1 + f_x^2 + f_y^2} =$

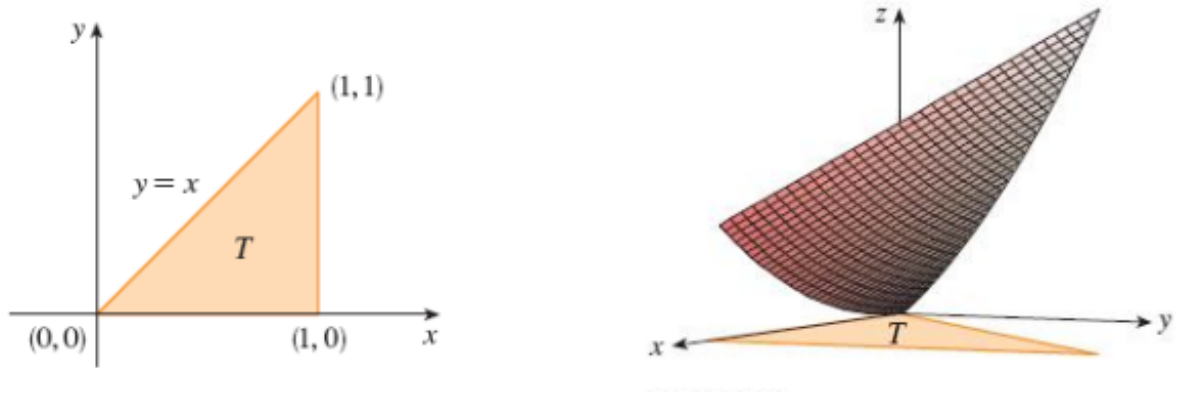


Figure 4.0.2: Figure for Example 4.0.4

$\sqrt{5 + 4x^2}$.

Observe that the region T can be expressed both as type I and type II regions. We can proceed with any choice. Here, we have considered T as a type I region.

So the area of the surface

$$= \int_T \int \sqrt{1 + f_x^2 + f_y^2} dA = \int_{x=0}^1 \int_{y=0}^x \sqrt{5 + 4x^2} dy dx.$$

The rest of the calculation is left to the reader as an exercise. $\square$

Example 4.0.5. Find the area of the part of the paraboloid $z = x^2 + y^2$ that lies under the plane $z = 9$. Figure 4.0.3 shows the surface in blue color.

Observe that the projection of the required portion of the paraboloid on the xy -plane will be the circular disk D (as shown in Figure 4.0.3) given by

$$D = \{(x, y) | 0 \leq x^2 + y^2 \leq 9\}.$$

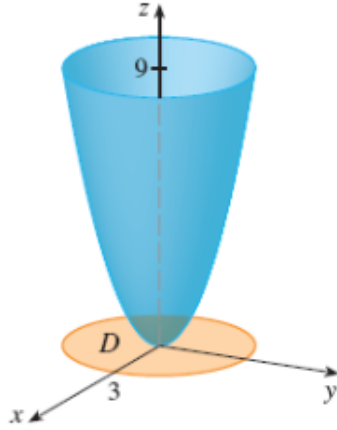


Figure 4.0.3: Figure for Example 4.0.5

Here $z = x^2 + y^2$. So $z_x = 2x$ and $z_y = 2y$. Hence the required surface area

$$= \int_D \int \sqrt{1 + 4(x^2 + y^2)} dA.$$

We could have worked with the region D expressed (as above) in terms of cartesian coordinates. But if we express D in terms of polar coordinates, our computation is going to be easier. In terms of polar coordinates, we have

$$D = \{(r, \theta) | 0 \leq r \leq 3, 0 \leq \theta \leq 2\pi\}.$$

Hence, the required surface area

$$= \int_{\theta=0}^{2\pi} \int_{r=0}^3 \sqrt{1 + 4r^2} r dr d\theta.$$

The rest of the calculation is left to the reader as an exercise. □

—————-END—————